

Bridge Developer Guide

Introduction, Supported Networks, Contracts, Core Concepts, and Fees

Document scope

A structured, developer-friendly reference based on the official BOT Chain Bridge documentation pages from Introduction through Fees. The source material has been paraphrased and reorganized for clarity; it is not a verbatim republication.

Official documentation	https://dev-docs.botchain.ai/docs/Bridge/
Coverage	Introduction → Supported Chains & Assets → Contract Addresses → Core Concepts → Fees
Primary asset	USDT
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1. Executive overview

BOT Chain Bridge is the official cross-chain transfer system for moving USDT between BOT Chain and supported external networks. The documented architecture uses chain-specific USDT liquidity vaults and a lock-and-release process rather than minting wrapped USDT.

Item	Officially documented position
Supported asset	USDT only
BOT Chain mainnet ID	677
BOT Chain testnet ID	968
External networks	Ethereum, BNB Smart Chain, Tron
Bridge-in fee	0 USDT bridge fee; source-network gas still applies
Bridge-out fee	0.1%, with a minimum fee of 1 USDT
Minimum bridge-out amount	Transfer amount must be greater than 10 USDT
Liquidity model	Lock and release using USDT vaults
Transfer authorization	Validator attestations and an on-chain signature threshold
Emergency control	BridgeRouter can be paused by governance

Important verification note

The Bridge landing page summary contains an inconsistent chain-ID statement, but the detailed Contract Addresses page identifies mainnet as Chain ID 677 and testnet as Chain ID 968.

Developers should verify production values against the current official chain configuration before deployment.

2. Introduction to the BOT Chain Bridge

The bridge connects BOT Chain with Ethereum, BNB Smart Chain, and Tron so that native USDT can move into and out of the BOT Chain ecosystem. The documentation describes BOT Chain as an EVM-compatible Layer 1 with approximately 0.75-second block times, Physical Finality, and low transaction costs.

Why the bridge is positioned as useful

- No bridge fee for deposits into BOT Chain.
- Predictable withdrawal pricing: 0.1% with a minimum fee of 1 USDT.
- Fast confirmation on the BOT Chain side because of its short block time and finality model.
- Cross-chain verification through validators or multi-signature-style attestations, with transfer evidence enforced on-chain.
- Coverage for ERC-20, BEP-20, and TRC-20 USDT.

Recommended reading path for implementers

1. Confirm supported origin and destination networks.
2. Verify all token and bridge contract addresses.

3. Understand lock/release, validator, finality, and replay-protection behavior.
4. Apply the correct fee and minimum-transfer rules in the user interface and backend.

3. Supported chains and assets

The official Bridge documentation currently limits transfers to USDT. The supported network matrix is:

Network	Chain identifier	USDT standard
BOT Chain	677	USDT on BOT Chain
Ethereum	1	ERC-20 USDT
BNB Smart Chain	56	BEP-20 USDT
Tron	Tron Mainnet (non-EVM)	TRC-20 USDT

Tron implementation note

Tron is non-EVM, so wallet connection, address validation, transaction construction, event monitoring, and explorer links must be handled separately from Ethereum-compatible chains.

4. Contract addresses

4.1 BOT Chain Mainnet

Contract	Address	Purpose
BridgeRouter	0xef8DC669ECa13E612b67Ff09478352E85bD6CC53	Entry point for bridge-in and bridge-out requests
USDT	0xaBabc7Ddc03e501d190C676BF3d92ef0e6e87a3C	USDT token contract on BOT Chain
Multicall3	0x47FA21f684bBAD707A53a0f9BE59F1422F46C265	Batch read/call helper

4.2 External networks

Network / Contract	Address	Role
Ethereum BridgeGateway	0x2945d3aF6f012e49f7421252b5fB57D1bb7E6Edd	Bridge counterpart on Ethereum
Ethereum USDT	0xdAC17F958D2ee523a2206206994597C13D831ec7	ERC-20 USDT
BNB Smart Chain BridgeGateway	0x3cd6fB6b0CDdD3610f0f4769AA7Bb686Cd4a4b55	Bridge counterpart on BNB Smart Chain
BNB Smart Chain USDT	0x55d398326f99059fF775485246999027B3197955	BEP-20 USDT
Tron BridgeGateway	TGhXbQpJBgC6bDp5jAexzeQPHEXXsx5f35	Bridge counterpart on Tron
Tron USDT	TR7NHqjeKQxGTCi8q8ZY4pL8otSzgijLj6t	TRC-20 USDT

4.3 BOT Chain Testnet

Contract	Address
BridgeRouter	0x6239404Aa276ba68486E2Fa40E90CDd36ff8ec3A
USDT (testnet)	0x75edC9335175Fc0552D51D48439F229c10420fe3

Shared USDT resource ID: 0xac589789ed8c9d2c61f17b13369864b5f181e58eba230a6ee4ec4c3e7750cd1d

Production safety rule

Never hard-code these addresses without environment separation. Maintain an audited chain registry keyed by chain ID, validate bytecode at startup, and block transactions when the connected network does not match the selected route.

5. Bridge contract functions

The documentation presents two deposit entry scenarios:

Function	Use case	Destination result
deposit(...)	Standard cross-chain USDT transfer	Recipient receives the destination-chain USDT amount according to bridge processing rules
depositWithBotGas(...)	Transfer into BOT Chain while also obtaining BOT for gas	Recipient receives USDT plus BOT; the BOT cost is deducted from the net USDT deposit amount

UX implication

For depositWithBotGas(...), the interface should show the quoted BOT allocation, USDT deduction, expected net USDT, price/slippage or quote-expiry rules if applicable, and the final destination amounts before the wallet confirmation.

6. Core concepts and security model

6.1 Lock and release liquidity model

The bridge uses native USDT liquidity on each supported chain. It does not rely on issuing a wrapped representation of USDT.

- Bridge in: USDT is locked in a source-chain TokenVault, and an equal quantity is released from the BOT Chain vault to the recipient.
- Bridge out: USDT is locked in the BOT Chain vault, and the destination-chain vault releases the corresponding amount after deducting the applicable withdrawal fee.

6.2 Validators and relayers

Validators independently monitor source-chain deposit events and sign attestations. Once enough signatures are collected, a relayer submits the aggregated proof on the destination chain. The on-chain BridgeValidator logic checks that the required signature threshold has been met before funds are released.

- Validators attest to events; the documentation states that they do not take custody of user funds.
- Relayers transport the aggregated proof to the destination contract.
- Release authorization is enforced by the smart contract rather than by a single off-chain operator.

6.3 Confirmations and finality

- BOT Chain: transactions are described as final once included because of Physical Finality, allowing near-immediate progression on the BOT Chain side.
- Ethereum, BNB Smart Chain, and Tron: the bridge waits for a chain-specific number of confirmations to reduce reorganization risk before validators attest.

6.4 Documented security controls

- Each transfer has a unique transferId and may be claimed only once, providing replay protection.
- Release requires a valid quorum of validator signatures checked on-chain.
- Vault liquidity is described as 1:1 backed by locked USDT.
- Governance can pause the BridgeRouter during an emergency.

Security due diligence beyond the documentation

Before integrating production funds, independently verify contract source code, upgradeability and admin controls, validator-set governance, quorum thresholds, vault solvency monitoring, pause/unpause authority, rate limits, incident procedures, and audit reports. Documentation alone is not a security audit.

7. Fees and transfer rules

Direction	Bridge fee	Rule
Into BOT Chain	0 USDT	No bridge fee; source-chain network gas still applies
Out of BOT Chain	0.1%, minimum 1 USDT	Fee is deducted from the transfer amount; transfer amount must be greater than 10 USDT

Fee calculation

Bridge-out fee = max(transfer amount × 0.001, 1 USDT)

Recipient amount = transfer amount - bridge-out fee

Requested amount	0.1% calculation	Applied fee	Expected recipient amount
11 USDT	0.011 USDT	1 USDT minimum	10 USDT
500 USDT	0.5 USDT	1 USDT minimum	499 USDT

1,000 USDT	1 USDT	1 USDT	999 USDT
10,000 USDT	10 USDT	10 USDT	9,990 USDT

Frontend validation

Reject bridge-out amounts of 10 USDT or less, display the bridge fee separately from gas, and show the exact expected destination amount before submission. Gas costs may change independently of the fixed bridge-fee formula.

8. Developer implementation checklist

- Create separate mainnet and testnet configuration files; default all development and QA to testnet.
- Validate the connected wallet network and destination address format before enabling submission.
- Use the official USDT decimals and contract metadata from the deployed token contract, not assumptions.
- Implement token approval checks and explain approval versus bridge submission as separate transactions.
- Generate or capture a stable client-side operation ID and store the on-chain transaction hash immediately.
- Monitor source-chain confirmations, validator processing, destination release, and final recipient balance.
- Display transfer status as resumable steps so a page refresh or device change does not lose progress.
- Protect against duplicate submission and duplicate claim attempts.
- Calculate fees deterministically and compare them with contract-returned or event values.
- Add route-specific handling for Tron because it is not EVM-compatible.
- Surface explorer links for both source and destination transactions.
- Create alerting for paused contracts, failed relays, low vault liquidity, delayed transfers, and contract-address changes.
- Perform contract ABI and bytecode verification before each release.
- Include clear user warnings that bridge transactions are irreversible once confirmed.

9. Documentation consistency notes

Item	Observed documentation	Recommended handling
BOT Chain chain ID	The Bridge landing summary references 968 in one sentence, while the detailed contract page states Mainnet 677 and Testnet 968.	Use 677 for mainnet and 968 for testnet only after confirming against the current official network configuration.
Supported-chains page	The Tron row is rendered	Treat Tron as Tron Mainnet /

formatting	awkwardly in the page text extraction.	non-EVM / TRC-20 USDT, and verify any additional numeric identifier directly with the bridge team.
Bridge URLs and branding	The documentation footer links to bridge.bohr.life, while current public ecosystem branding may use botchain.ai domains.	Use only officially announced production URLs and protect users from look-alike domains.
Operational parameters	Exact external-chain confirmation counts and validator quorum values are not listed on these pages.	Read the live contracts or obtain official operator specifications before implementing SLA estimates.

10. Official source pages

Bridge section: <https://dev-docs.botchain.ai/docs/Bridge/>

Introduction: <https://dev-docs.botchain.ai/docs/Bridge/introduction/>

Supported Chains & Assets: <https://dev-docs.botchain.ai/docs/Bridge/supported-chains/>

Contract Addresses: <https://dev-docs.botchain.ai/docs/Bridge/contract-addresses/>

Core Concepts: <https://dev-docs.botchain.ai/docs/Bridge/core-concepts/>

Fees: <https://dev-docs.botchain.ai/docs/Bridge/fees/>

Source note: This guide is a paraphrased and reorganized technical reference derived from the official BOT Chain Developer Documentation as accessed on 19 July 2026. Contract addresses, fees, and operational parameters can change; always re-check the live official documentation and deployed contracts before production use.